

Exam 3: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (10 pts) Determine the total change in the fuel volume given: $\alpha_{th}=10 \times 10^{-6}$, fission rate = 8×10^{13} fissions/cm³-s, $T_f=1500$ K, $T_{ref}=300$ K, $\Delta\rho_0=0.015$, $B_D=5$ MWD/kgU, $\rho(UO_2)=10.97$ g/cc, $t=300$ days.
2. (8 pts) What is the total creep in a RXA zirconium cladding given a von Mises stress of 60 MPa, a temperature of 600 K, a LHR of 250 W/cm, and a time of 200 days? $A_0 = 3.14 \times 10^{24}$; $G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T$; $n = 5$; $Q = 2.7 \times 10^5$, $C_0=1.654E-24$; $C_1=0.85$; $C_2=1$
3. (8 pts) Two fuel pellets are irradiated for 250 days at 1400 K at a fission rate of 2×10^{13} fissions/cm³-s. One fuel sample has a grain size of 5 microns, and the other has a grain size of 40 microns. What fraction of fission gas is released in each?
4. (4 pts) What is the typical charge state of U in UO₂? What can cause it to change?
5. (5 pts) How does the oxygen concentration change across the fuel pellet? How does Mo impact this?
6. (3 pts) Name three properties that vary as a function of stoichiometry in UO₂.
7. (10 pts) What are the five types of fission products that form in the fuel? List a role each plays in fuel performance.
8. (8 pts) List and describe the three stages of fission gas release.
9. (5 pts) What is creep? What is one mechanism of creep (name and physical description)?
10. (8 pts) What is the high burnup structure? Why does it form? What are some performance aspects of this structure?
11. (6 pts) Describe the irradiation growth of hcp Zr due to point defects.
12. (8 pts) Describe how creep impacts the Zr cladding over time in a fuel rod.
13. (12 pts) What are the four conditions that must be met for SCC? Briefly describe how each is met in PCI.
14. (5 pts) What role does iodine play in the corrosion of Zr cladding?

Equations for NE533

$$Q = E_f \times N_f \times \sigma_f \times \phi$$

$$T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T_0 - T_s = \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k}$$

$$T_s - T_{IC} = \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}}$$

$$T_{IC} - T_{OC} = \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}}$$

$$T_{OC} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}}$$

$$h_{gap/clad} = \frac{k_{gap/clad}}{t_{gap/clad}}$$

$$LHR \left(\frac{z}{Z_0} \right) = LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right]$$

$$T_{cool}(z) - T_{cool}^{in} = \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \times$$

$$\left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\}$$

$$\sigma_\theta = \frac{pR}{\delta} \quad \sigma_z = \frac{pR}{2\delta} \quad \sigma_r = -\frac{p}{2}$$

$$\sigma_r(r) = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_\theta(r) = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_z(r) = p \frac{1}{(R_o/R_i)^2 - 1}$$

$$\sigma = -E\Delta T\alpha$$

$$\sigma_r(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_\theta(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_z(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_r(\eta) = -\sigma^*(1 - \eta^2)$$

$$\sigma_\theta(\eta) = -\sigma^*(1 - 3\eta^2)$$

$$\sigma_z(\eta) = -2\sigma^*(1 - 2\eta^2)$$

$$\sigma^* = \frac{\alpha E(T_0 - T_s)}{4(1 - \nu)}$$

$$\Delta\delta_{gap} = \bar{R}_c\alpha_c(\bar{T}_c - T_{ref}) - R_f\alpha_f(\bar{T}_f - T_{ref})$$

$$D(t)^2 - D_0^2 = 2M_{GB}\gamma_{GB}t$$

$$M_{GB}(T) = M_0 e^{(-\frac{Q}{k_B T})}$$

$$\frac{dD}{dt} = k \left(\frac{1}{D} - \frac{1}{D_m} \right)$$

$$\epsilon_D = \Delta\rho_0 \left(e^{\frac{\beta I n_{0.01}}{C_D \beta_D}} - 1 \right)$$

$$\beta = \frac{\dot{F}t}{N_U}$$

Post-irradiation Annealing

$$\tau < \pi^{-2} \rightarrow f = 6\sqrt{\frac{Dt}{\pi a^2}} - 3\frac{Dt}{a^2}$$

$$\tau \geq \pi^{-2} \rightarrow f = 1 - \frac{6}{\pi^2} \exp(-\pi^2 \frac{Dt}{a^2})$$

In-pile gas release

$$\tau < \pi^{-2} \rightarrow f = 4\sqrt{\frac{Dt}{\pi a^2}} - \frac{3}{2} \frac{Dt}{a^2}$$

$$\tau \geq \pi^{-2} \rightarrow f = 1 - \frac{0.662}{\frac{Dt}{a^2}} \left(1 - 0.92 \exp(-\pi^2 \frac{Dt}{a^2}) \right)$$

$$D = D_1 + D_2 + D_3 \text{ cm}^2/\text{s}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.03}{k_B T}\right)$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{k_B T}\right) \sqrt{\dot{F}}$$

$$D_3 = 2.0 \times 10^{-30} \dot{F}$$

$$\epsilon_{SFP} = 5.577 \times 10^{-2} \rho \beta$$

$$\epsilon_{GFP} = 1.96 \times 10^{-28} \rho \beta (2800 - T)^{11.73} \times$$

$$e^{-0.0162(2800-T)} e^{-17.8\rho\beta}$$

$$\dot{\epsilon}_{th} = A_0 \left(\frac{\sigma}{G} \right)^n e^{(\frac{-Q}{RT})}$$

$$\dot{\epsilon}_{ir} = C_0 \Phi^{C_1} \sigma^{C_2}$$

$$\Phi = 3E11 \times LHR$$